



In Linux, SSHFS or the SSH File System is used to mount and interact with directories and files that are located on a remote server. Let's take a quick look at how to use this file system.

In large software organisations, it is very common to have separate development and build machines. Often, time is spent copying changes from the dev machine to the build machine. This becomes annoying when there are numerous files. Wouldn't it be great if we could use a remote directory as a local one? It is very easy to do this in GNU/Linux and this article tells you how.

Overview

SSHFS is a fuse file system based on the SSH protocol and one of its big advantages is that it has to be installed on the client side only. No installation or configuration is required on the server side. The only requirement is that the openSSH server should be installed on the server side. In this article, we'll discuss the usage and various features of SSHFS.

Installation

Installing SSHFS is pretty straightforward – it's just like installing other packages. If you are using a Debian based GNU/Linux distro, such as Debian, Mint or Ubuntu, then execute the following commands:

```
$ sudo apt-get update
$ sudo apt-get install sshfs
```

If you are using RPM based GNU/Linux, such as Red Hat or CentOS, then execute the following commands:

```
$ sudo yum update
$ sudo yum install fuse-sshfs
```

Mounting a remote directory

The syntax of the SSHFS command is:

```
sshfs [user@]host:[directory] mountpoint [options]
```

To mount a remote directory in GNU/Linux, execute the following command:

```
$ sshfs jarvis@192.168.10.20:/home/jarvis/src /mnt
```

In the above command:

- `jarvis` is a user
- `192.168.10.20` is a remote host IP address
- `/home/jarvis/src` is a directory present on the remote host
- `/mnt` is a mount point

Configuring a passwordless SSH mount

In the above method, each time you mount a directory, it will ask for credentials. We can avoid this by configuring a passwordless SSH between these two machines. To do this, execute the following commands:

```
$ ssh-keygen -t rsa
$ ssh-copy-id -i ~/.ssh/id_rsa.pub jarvis@192.168.10.20
```

In the above example:

- The first command creates RSA key pairs
- The second command copies the RSA public key to the remote host